

Fabian Lehmann

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Education

MSc Artificial Intelligence in Medicine University of Bern	2025 – 2027
BSc Artificial Intelligence & Machine Learning Lucerne University of Applied Sciences and Arts (HSLU)	2022 – 2025
Automation Technician (Automatiker EFZ) login Berufsbildung AG (RhB)	2017 – 2021

Projects

Medical Image Segmentation (Bachelor Thesis)

- Developed 2D and 3D models (UNet, ViT, Mamba) for cardiac MRI segmentation on the ACDC dataset
- Applied semi-supervised learning using pseudo and sparse labels
- Achieved Dice scores of 89.3% (2D) and 85.5% (3D)

BELKA Kaggle Challenge (Drug Discovery)

- Built end-to-end ML pipeline for protein–molecule binding prediction on 295M+ samples, including SMILES preprocessing, feature engineering, and dataset restructuring
- Developed and evaluated tree-based and transformer models (XGBoost, ChemBERTa) for large-scale, highly imbalanced classification, leveraging transfer learning to improve generalization
- Processed 50GB+ datasets using Dask and implemented efficient sampling strategies to enable training under limited compute resources

Experience

Programming Instructor (Part-Time) CodeCamp	2022 – Present
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- Teach and support students in problem-solving and programming fundamentals

Scout Leader (Volunteer) BESJ – (Bund Evangelisch Schweizerischer Jungscharen)	2016 – 2024
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- Progressed from Scout Leader to Material Manager to Team Leader
- Organized and led camps and events, including team coordination and logistics management

Technical Skills

Languages: Python, C/C++, R

ML/AI: PyTorch, Scikit-learn, XGBoost, Transformers, Time-Series Analysis, NLP, Computer Vision

Data: Pandas, NumPy, Matplotlib, Jupyter

Tools: Git, Docker, Weights & Biases, Linux

Languages

German (native), English (fluent)

Additional

Swiss Military Service – Reconnaissance Specialist (Aufklärer) (2022, 6 months)

Humanitarian Community Service in East Asia (2021–2022)